

Project Logbook

Date of Activity: Week ending 16/11/25

Summary of Work: Made a Data collection plan based on my Gantt chart and deliverables for the project. Started looking into datasets of images of plants, originally searching for specific weeds like stinging nettles, thorns, thistles, and dandelions. However, found a lack of available pre-made datasets. Decided it would be better to pivot to a comprehensive leaf dataset. Located the "UK Leaf Dataset" publicly available on Kaggle.

Actions Arising: Begin data preparation and set up the Python/TensorFlow environment for model development.

Date of Activity: Week ending 23/11/25

Summary of Work: Focused on Dataset Preparation and Model Selection. Successfully loaded the Kaggle UK Leaf Dataset. The dataset consists of 340 source images across 7 specific classes: Maple, Nettle, Hogweed, Blackberry, Blackthorn, Oak, and Laurel. Divided the images into a training set of 238 images, a validation set of 68 images, and a testing set of 34 images. Pre-processed the images by resizing them to a standard dimension of 224x224 to ensure consistency.

Actions Arising: Write the base code to compile and train candidate models.

Date of Activity: Week ending 30/11/25

Summary of Work: Began Model Training. Selected multiple machine learning models for comparison, including a convolutional neural network and lightweight pre-trained models like MobileNet and EfficientNet. Progress is moving much faster than the Gantt chart predicted; multiple lightweight machine learning models have been successfully built and trained using Python in Google Colab, putting the project ahead of schedule.

Actions Arising: Prepare documentation for Review 2.

Date of Activity: Week ending 07/12/25

Summary of Work: Prepared for the Review 2 Meeting. Completed the Impact Reflection and finalized the Review of Key References. Showed my supervisor the early model training results, confirming we are well ahead of the scheduled "Testing and Validation" track for the software side.

Actions Arising: Take on feedback from Review 2. Begin Model Evaluation and Optimisation after the break.

Date of Activity: Week ending 18/01/26

Summary of Work: Evaluated the trained models to determine their suitability for real-time use based on overall accuracy, precision, and inference speed. Began the "Model Optimisation" phase to convert the models into a format suitable for edge devices.

Actions Arising: Procure the physical Raspberry Pi hardware and begin the "Model Integration" phase.

Date of Activity: Week ending 01/02/26

Summary of Work: Started physical hardware setup. Flashed the Raspberry Pi OS onto the SD card, set up VNC viewer for headless access, and transferred the .tflite model over to the Pi. Began writing the Python script to handle inference. Started noticing that hardware integration is going to be much slower and more tedious than the cloud-based ML development.

Actions Arising: Install TensorFlow Lite runtime and OpenCV on the Raspberry Pi to test the script.

Date of Activity: Week ending 15/02/26

Summary of Work: Hit a major roadblock causing significant delays in the "Model Integration" schedule. The Python script crashed repeatedly due to a TensorFlow version mismatch resulting in FULLY_CONNECTED version 12 optimization errors on the Pi's tflite-runtime library. Spent hours troubleshooting the compatibility between Google Colab's export tools and the older edge architecture.

Actions Arising: Find a workaround for the version mismatch without having to completely rebuild the Raspberry Pi operating system.

Date of Activity: Week ending 01/03/26

Summary of Work: Resolved the version mismatch delay. Instead of upgrading the Pi, I went back to Colab and exported an uncompressed .tflite model, which successfully resolved the optimization errors. The AI model successfully loaded on the Pi without crashing. Connected the physical camera module to attempt real-time testing.

Actions Arising: Address camera loading errors and meet with my supervisor regarding model generalization.

Date of Activity: Week ending 08/03/26

Summary of Work: To improve model generalization, it was proposed that I implement data augmentation by my professor. I integrated "on-the-fly" preprocessing directly into the training pipeline, applying horizontal and vertical flips, rotations between -27 and +27 degrees, horizontal and vertical shear, and brightness adjustments. This

successfully created a robust training set of 816 images. Retrained and re-exported the final optimized models.

Actions Arising: Return to the Raspberry Pi to fix the camera software conflicts.

Date of Activity: Week ending 15/03/26

Summary of Work: Addressed the massive hardware/software conflict preventing the camera from feeding images to the model. Successfully overcame the software conflicts between the modern Raspberry Pi OS libcamera architecture and OpenCV's memory allocation. A live video feed was successfully established and integrated into the Python environment. The "Smart Glasses" prototype can now successfully capture physical images and run real-time, on-device inferences.

Actions Arising: Finalize the Skeleton Report Structure and structured abstract for the Review 3 meeting.